Roll No. | | | | | | | |

End Semester Examination 2025

Name of the Course: B. Tech (Civil Engineering) — Semester: IV
Name of the Paper: Hydraulics and Hydraulic Machines — Paper Code: TCE 401
Time: 3 Hours — Maximum Marks: 100

Note:
All questions are compulsory
Answer any two sub questions among a, b & c in each main questions
Total marks in each question are twenty
Each question carries 10 marks

Q1 **(20 marks)**

(a) State Rayleigh method of dimensional analysis and explain procedure for the analysis.

(b) Determine the most efficient section of a trapezoidal channel with side slopes 1V to 2 H carrying a discharge of 11.25 m^3/s with a velocity of 0.75 m/s. What should be the bed slope of the channel? Take Manning's n = 0.025. **CO1**

(c) A rectangular channel is 2.50 m wide and carries a flow of 2.75 m^3/s at a depth of 0.9 m. A contraction of the channel width is proposed at a certain section. Find the smallest allowable contracted width that will not affect the upstream flow conditions.

Q2 **(20 marks)**

(a) Discuss the differential equation for water surface slope in a Gradually Varied Flow. List all the assumptions.

(b) In a rectangular channel 12 m wide and 3.6 m deep, water is flowing with a velocity of 1.2 m/s. the bed slope of channel is 1 in 4000. If the flow of water through the channel is regulated in such a way that energy line is having a slope of 0.00004, find the rate of change of depth of water in the channel. **CO2**

(c) Water flows in a rectangular channel 0.60 m wide at a depth of 0.18 m and Froude number 2.75. Determine the specific energy head; the critical depth; the sequent depth and head lost due to hydraulic jump.

Q3 **(20 marks)**

(a) Derive an expression for the lift produced on a rotating cylinder placed in a uniform flow field such that the axis of the cylinder is perpendicular to the direction of flow.

(b) A thin plate is moving in still atmospheric air at a velocity of 5 m/s. The length of the plate is 0.6 m and width 0.5 m. Calculate the thickness of the boundary layer at the end of the plate and drag force on one side of the plate. Take density of air as 1.24 kg/m^3 and kinematic viscosity 0.15 stokes. **CO3**

(c) A passenger car with frontal projected area of 1.5 m^2 travels at 56 km/hour. Determine the power required to overcome wind resistance if the drag coefficient of the car is 0.40. For the same power expended in overcoming resistance, what percentage change in speed of the car is possible if drag coefficient is reduced to 0.32 by streamlining the car body? Take density of the air = 1.2 kg/m^3.

Q4 **(20 marks)**

(a) Derive an expression for the force exerted by a jet of water on a fixed vertical plate in the direction of the jet.

(b) Explain with sketches the working of a single stage centrifugal pump. **CO4**

(c) Calculate vane angle at the inlet of a centrifugal pump impeller having 200 mm diameter at inlet and 400 mm diameter at outlet. The impeller vanes are set back at an angle of 45^0 to the outer rim, and the entry of the pump is radial. The pump runs at 1000 r.p.m. and the velocity of flow through the impeller is constant at 3 m/s. Also calculate the velocity as well as direction of the water at outlet.

Q5 **(20 marks)**

(a) Describe with the help of sketch, the working principle of Francis turbine.

(b) Write a short note on the factors influencing the performance of turbine.

(c) A Pelton wheel having semicircular buckets and working under a head of 140 m, is running at 600 r.p.m. The discharge through the nozzle is 500 litres/s and diameter of the wheel is 600 mm. Find the power available at the nozzle and hydraulic efficiency of the wheel, if coefficient of velocity is 0.98. **CO5**